

Given a finite number of finite sets, $A_1, A_2, \dots, A_n$, the number of elements in the union $A_1 \cup A_2 \cup \dots \cup A_n$ is

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|,$$

where the first sum is over all i , the second sum is over all pairs i, j with $i < j$, the third sum is over all triples i, j, k with $i < j < k$, and so forth.

The number of derangements of $n \geq 1$ ordered symbol is

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^n \frac{1}{n!} \right).$$

Let $a_0, a_1, a_2, \dots$ be a sequence of real numbers. The function

$$f(x) = a_0 + a_1x + a_2x^2 + \dots = \sum_{i=0}^{\infty} a_i x^i$$

is called the generating function for the given sequence.

4.1.2 Extended Binomial coefficient

$$\begin{aligned} \binom{-n}{r} &= \frac{(-n)(-n-1)(-n-2)\dots(-n-r+1)}{r!} \\ &= \frac{(-1)^r (n)(n+1)(n+2)\dots(n+r-1)}{r!} \\ &= \frac{(-1)^n (n+r-1)!}{(n-1)!r!} = (-1)^r \binom{n+r-1}{r}. \end{aligned}$$

For all $m, n \in \mathbb{Z}^+, a \in \mathbb{R}$,

- (1) $(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$
- (2) $(1+ax)^n = \binom{n}{0} + \binom{n}{1}ax + \binom{n}{2}a^2x^2 + \dots + \binom{n}{n}a^nx^n$
- (3) $(1+x^m)^n = \binom{n}{0} + \binom{n}{1}x^m + \binom{n}{2}x^{2m} + \dots + \binom{n}{n}x^{nm}$
- (4) $(1-x^{n+1})/(1-x) = 1 + x + x^2 + \dots + x^n$
- (5) $1/(1-x) = 1 + x + x^2 + x^3 + \dots = \sum_{i=0}^{\infty} x^i$
- (6) $1/(1-ax) = 1 + (ax) + (ax)^2 + (ax)^3 + \dots$
 $= \sum_{i=0}^{\infty} (ax)^i = \sum_{i=0}^{\infty} a^i x^i$
 $= 1 + ax + a^2x^2 + a^3x^3 + \dots$

$$\begin{aligned}
 (7) \quad 1/(1+x)^n &= \binom{-n}{0} + \binom{-n}{1}x + \binom{-n}{2}x^2 + \dots \\
 &= \sum_{i=0}^{\infty} \binom{-n}{i} x^i \\
 &= 1 + (-1) \binom{n+1-1}{1} x + (-1)^2 \binom{n+2-1}{2} x^2 + \dots \\
 &= \sum_{i=0}^{\infty} (-1)^i \binom{n+i-1}{i} x^i
 \end{aligned}$$

$$\begin{aligned}
 (8) \quad 1/(1-x)^n &= \binom{-n}{0} + \binom{-n}{1}(-x) + \binom{-n}{2}(-x)^2 + \dots \\
 &= \sum_{i=0}^{\infty} \binom{-n}{i} (-x)^i \\
 &= 1 + (-1) \binom{n+1-1}{1} (-x) + (-1)^2 \binom{n+2-1}{2} (-x)^2 + \dots \\
 &= \sum_{i=0}^{\infty} \binom{n+i-1}{i} x^i
 \end{aligned}$$

If $f(x) = \sum_{i=0}^{\infty} a_i x^i$, $g(x) = \sum_{i=0}^{\infty} b_i x^i$, and $h(x) = f(x)g(x)$, then

$h(x) = \sum_{i=0}^{\infty} c_i x^i$, where for all $k \geq 0$,

$$c_k = a_0 b_k + a_1 b_{k-1} + \dots + a_{k-1} b_1 + a_k b_0 = \sum_{j=0}^k a_j b_{k-j}.$$

Objects Are Distinct	Containers Are Distinct	Some Container(s) May Be Empty	Number of Distributions
No	Yes	Yes	$\binom{n+m-1}{m}$
No	No	Yes	(1) $p(m)$, for $n = m$ (2) $p(m, 1) + p(m, 2) + \dots + p(m, n)$, for $n < m$
No	Yes	No	$\binom{n+(m-n)-1}{(m-n)} = \binom{m-1}{m-n} = \binom{m-1}{n-1}$
No	No	No	$p(m, n)$

Consider the nonhomogeneous first-order relation

$$a_n + C_1 a_{n-1} = k r^n,$$

where k is a constant and $n \in \mathbb{Z}^+$. If r^n is not a solution of the associated homogeneous relation

$$a_n + C_1 a_{n-1} = 0,$$

then $a_n^{(p)} = Ar^n$, where A is a constant. When r^n is a solution of the associated homogeneous relation, then $a_n^{(p)} = Bnr^n$, for B a constant.

Now consider the case of the nonhomogeneous second-order relation

$$a_n + C_1a_{n-1} + C_2a_{n-2} = kr^n,$$

Where k is a constant. Here we find that

- (a) $a_n^{(p)} = Ar^n$, for A a constant, if r^n is not a solution of the associated homogeneous relation;
- (b) $a_n^{(p)} = Bnr^n$, where B is a constant, if $a_n^{(h)} = c_1r^n + c_2r^n$, where $r_1 \neq r$; and
- (c) $a_n^{(p)} = Cn^2r^n$, for C a constant, when $a_n^{(h)} = (c_1 + c_2n)r^n$.

Given a linear nonhomogeneous recurrence relation (with constant coefficients) of the form $C_0a_n + C_1a_{n-1} + C_2a_{n-2} + \dots + C_ka_{n-k} = f(n)$, where $C_0 \neq 0$ and $C_k \neq 0$, let $a_n^{(h)}$ denote the homogeneous part of the solution a_n .

	$a_n^{(p)}$
c , a constant	A , a constant
n	$A_1n + A_0$
n^2	$A_2n^2 + A_1n + A_0$
n^t , $t \in \mathbb{Z}^+$	$A_tn^t + A_{t-1}n^{t-1} + \dots + A_1n + A_0$
r^n , $r \in \mathbb{R}$	Ar^n
$\sin \theta n$	$A \sin \theta n + B \cos \theta n$
$\cos \theta n$	$A \sin \theta n + B \cos \theta n$
$n^t r^n$	$r^n (A_tn^t + A_{t-1}n^{t-1} + \dots + A_1n + A_0)$
$r^n \sin \theta n$	$Ar^n \sin \theta n + Br^n \cos \theta n$
$r^n \cos \theta n$	$Ar^n \sin \theta n + Br^n \cos \theta n$

